CH 6 — Chain Rule

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6.1 Chain Rule

Recall we use the chain rule to take derivatives of composite functions.

Consider

$$T(t) = f(x(t))$$

where $x(t)$ represents the position of an ant crowling along a log and $T(t)$ represents the remperature of the log as the ant moves along.

By the chain rule the change in the temperature can be obtained

$$T'(t) = f'(x(t)) \cdot x'(t)$$

In 2D, if we allow the ant to move in any direction, that is $f(x, y)$ is the temperature at the point (x, y) and $\vec{x}(t) = (x(t), y(t))$ represents the position of the ant as a function of time.

Then,

$$T(t) = f(x(t), y(t))$$

represents the temperature experienced by the ant as it moves along its path.

Let t change by a small amount Δt, then $\begin{cases} \Delta x = x(t+\Delta t) - x(t) \\ \Delta y = y(t+\Delta t) - y(t) \end{cases}$.

By the increment form of linear approximation,

$$\Delta T \approx \frac{\partial T}{\partial x}\Delta x + \frac{\partial T}{\partial y}\Delta y$$

$$\lim_{\Delta t \to 0} \frac{\Delta T}{\Delta t} \approx \lim_{\Delta t \to 0} \left[\frac{\partial T}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial T}{\partial y}\frac{\Delta y}{\Delta t} \right]$$

$$\frac{\mathrm{d}T}{\mathrm{d}t} \approx \frac{\partial T}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial T}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

If $T(x, y)$ is differentiable, we can show that the error in the linear approximation gives 0

> 📘 **Definition — Chain Rule**
>
> Let $T(t) = f(x(t), y(t))$ and $a = x(t_0), b = y(t_0)$. If f is differentiable at (a, b) and both $x'(t_0)$ and $y'(t_0)$ exist. Then $T'(t)$ exists and given by
>
> $$T'(t_0) = f_x(a, b) \cdot x'(t_0) + f_y(a, b) \cdot y'(t_0)$$

Example:

Given $T(x, y) = x^2 + 2y^2, x(t) = e^t \sin t, y(t) = \cos t$, find $T'(t)$

$T'(t) = \left[e^{2t}\sin^2 t + 2\cos^2 t\right]' = 2te^{2t}\sin^2 t + 2e^{2t}\sin t\cos t - 4\cos t \sin t$

Alternatively,

$$T'(t) = 2(e^t \sin t)(e^t \sin t + e^t \cos t) + 4(\cos t)(-\sin t) = 2e^{2t} \sin^2 t(\sin t + \cos t) - 4 \cos t \sin t$$

6.2 Generalization to $\mathbb{R}^n$

> 📘 **Definition — Generalization of Chain Rule**
>
> The definition of a differentiable fucntion and all the theorems on differentiablility generalize to $\mathbb{R}^n$
>
> Let $\vec{x}(t) \in \mathbb{R}^n, T(t) = f(\vec{x}(t))$, we have
>
> $$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mathrm{d}f(\vec{x}(t))}{\mathrm{d}t} = \boldsymbol{\nabla} f(\vec{x}(t)) \cdot \frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t}$$

Example

Let $f(x, y, z)$ be a differentiable function and $g(t) = f(x(t), y(t), z(t))$

where $x(t) = t, y(t) = t^2, z(t) = t^3$. If $\boldsymbol{\nabla} f(1, 1, 1) = \left(2, \frac{1}{2}, 1\right)$, find $g'(1)$

$$g'(t) = \boldsymbol{\nabla} f(\vec{x}(t)) \cdot \tfrac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t} \overset{t=1}{\Longrightarrow} g'(1) = \boldsymbol{\nabla} f(\vec{x}(1)) \cdot \tfrac{\mathrm{d}\vec{x}(1)}{\mathrm{d}t} = \boldsymbol{\nabla} f(1, 1, 1) \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{1}{2} \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 6$$